

SIMATS ENGINEERING

MACHINE LEARNING – CO3

MCQ QUESTIONS WITH CORRECT OPTIONS & EXPLANATIONS

Assessment Tool 3 / 50 Multiple Choice Questions

1. Bayes' Theorem is a fundamental concept in probability theory. Which of the following best describes its primary purpose in updating prior beliefs when new evidence is observed in a probabilistic model?

- a) To compute only prior probabilities
- b) To update prior beliefs using new evidence
- c) To eliminate irrelevant features
- d) To maximize classification accuracy

Correct Answer: B

Explanation: Bayes' theorem updates prior beliefs using new evidence.

Example: Example: Before a medical test, the disease probability is a prior. A positive test provides evidence and updates that probability.

2. In the Naïve Bayes classifier, a strong assumption is made about the relationship between features. Which of the following statements correctly describes this assumption in practical machine learning applications?

- a) Features are completely dependent on each other
- b) Features are conditionally independent given the class
- c) Features must be continuous
- d) Features are always correlated

Correct Answer: B

Explanation: Naïve Bayes assumes features are conditionally independent given the class.

Example: Example: In spam detection, word features are treated as conditionally independent given the class Spam or Not Spam.

3. Maximum Likelihood Estimation is widely used for parameter estimation in probabilistic models. Which of the following options correctly explains the objective of MLE when fitting a model to observed data?

- a) Maximize prior probability
- b) Minimize hypothesis space
- c) Maximize likelihood of observed data
- d) Reduce computational complexity

Correct Answer: C

Explanation: MLE chooses parameter values that maximize the likelihood of the observed data.

Example: Example: Estimate the mean of student marks so the observed marks are as likely as possible under the model.

4. The Minimum Description Length principle is often used in model selection. Which of the following best explains how MDL balances model complexity and goodness of fit in learning algorithms?

- a) Prefers highly complex models always
- b) Prefers simplest model that explains data well
- c) Ignores data noise completely
- d) Focuses only on training accuracy

Correct Answer: B

Explanation: MDL prefers the simplest model that explains the data well.

Example: Example: Choose a simple model instead of a complicated model that only memorizes training data.

5. The Bayes Optimal Classifier is considered theoretically ideal in classification tasks. Which of the following statements best describes how it achieves the minimum possible classification error?

- a) By selecting random hypotheses
- b) By averaging predictions over all hypotheses
- c) By using only prior probabilities
- d) By minimizing dataset size

Correct Answer: B

Explanation: The Bayes Optimal Classifier averages predictions over hypotheses according to their posterior probabilities.

Example: Example: Several possible classifiers contribute to the final prediction according to how probable each hypothesis is.

6. The Gibbs Algorithm is an important concept in Bayesian learning. Which of the following options best explains how it selects a hypothesis from the hypothesis space during classification?

- a) Selects hypothesis randomly from consistent set
- b) Selects hypothesis with minimum loss only
- c) Selects fixed predefined hypothesis
- d) Ignores hypothesis space

Correct Answer: A

Explanation: Gibbs sampling selects a hypothesis randomly according to the relevant

posterior/consistent hypothesis distribution.

Example: Example: If several hypotheses are consistent with the training data, the algorithm can randomly select one.

7. Naïve Bayes classifiers are often used in text classification tasks. Which of the following conditions ensures that the classifier performs effectively despite its simplifying assumptions?

- a) Strong dependence between features
- b) Large amount of labeled data
- c) Features being conditionally independent
- d) Absence of noise in dataset

Correct Answer: C

Explanation: Naïve Bayes performs well when its conditional-independence assumption is approximately satisfied.

Example: Example: Word features in a text classifier may be approximately conditionally independent given the class.

8. Bayesian Belief Networks are graphical models that represent probabilistic relationships. Which of the following best describes the role of nodes and edges in such networks?

- a) Nodes represent variables and edges represent dependencies
- b) Nodes represent datasets and edges represent features
- c) Nodes represent outputs only
- d) Edges represent noise in data

Correct Answer: A

Explanation: Nodes represent variables and directed edges represent probabilistic dependencies.

Example: Example: Disease → Symptoms shows that symptoms depend probabilistically on disease.

9. The Expectation-Maximization algorithm is commonly used when dealing with incomplete data. Which of the following best explains the role of the Expectation step in this iterative algorithm?

- a) Updates model parameters
- b) Computes expected values of hidden variables
- c) Removes missing data
- d) Sorts the dataset

Correct Answer: B

Explanation: The E-step computes expected values or responsibilities for hidden variables.

Example: Example: In clustering, it estimates how likely each data point belongs to each cluster.

10. In the EM algorithm, the Maximization step updates certain parameters. Which of the following correctly describes what is updated during this step to improve model likelihood?

- a) Hidden variables only
- b) Observed data values
- c) Model parameters maximizing likelihood
- d) Training dataset size

Correct Answer: C

Explanation: The M-step updates model parameters to maximize the expected likelihood.

Example: Example: In a Gaussian mixture, cluster means and variances are updated.

11. Probability learning involves estimating probability distributions from data. Which of the following best explains the goal of probability learning in the context of machine learning models?

- a) To remove all uncertainty
- b) To estimate underlying probability distributions
- c) To reduce number of features
- d) To simplify model structure

Correct Answer: B

Explanation: Probability learning estimates the underlying probability distribution represented by the data.

Example: Example: Estimate the probability of rain from historical weather observations.

12. Sample complexity is an important concept in learning theory. Which of the following best describes what sample complexity indicates about a learning algorithm's performance?

- a) Amount of computation required
- b) Number of samples needed for learning
- c) Model accuracy only
- d) Memory consumption

Correct Answer: B

Explanation: Sample complexity indicates how many training examples are needed to learn reliably to a desired level.

Example: Example: A classifier may require thousands of labeled examples to reach a target accuracy.

13. A finite hypothesis space contains a limited number of possible hypotheses. Which of the following best explains how this limitation impacts the learning process and generalization ability?

- a) Improves flexibility always
- b) Restricts possible models considered
- c) Eliminates need for data
- d) Guarantees perfect accuracy

Correct Answer: B

Explanation: A finite hypothesis space restricts the number of models the learner can consider.

Example: Example: A learner may be allowed to choose only from four predefined rules.

14. In contrast to finite hypothesis spaces, infinite hypothesis spaces allow more flexibility. Which of the following best describes the characteristics of such hypothesis spaces in learning models?

- a) Limited to discrete functions
- b) Allows continuous and complex models
- c) Contains no hypotheses
- d) Simplifies computation always

Correct Answer: B

Explanation: An infinite hypothesis space can contain continuous or arbitrarily many possible models.

Example: Example: A threshold can be any real value such as 72.1234.

15. The mistake bound model is used to evaluate learning algorithms. Which of the following best describes what is measured by the mistake bound in online learning scenarios?

- a) Training time
- b) Maximum number of mistakes made
- c) Memory usage
- d) Dataset size

Correct Answer: B

Explanation: A mistake bound measures the maximum number of prediction mistakes made by an online learner.

Example: Example: An algorithm may have a guarantee that it makes no more than a certain number of mistakes.

16. Prior probability is a key concept in Bayesian inference. Which of the following best describes what prior probability represents before observing any new evidence?

- a) Probability after evidence
- b) Initial belief before observing data
- c) Conditional probability only
- d) Joint probability

Correct Answer: B

Explanation: Prior probability represents the initial belief before observing new evidence.

Example: Example: The initial probability that it will rain before checking today's weather.

17. Posterior probability is calculated after incorporating evidence. Which of the following best describes how posterior probability is derived using Bayes' Theorem?

- a) Using only likelihood
- b) Using prior and likelihood with evidence
- c) Ignoring prior information
- d) Using random sampling

Correct Answer: B

Explanation: Posterior probability combines prior information and likelihood, normalized by the evidence.

Example: Example: A disease prior is updated using the likelihood of a positive test.

18. Likelihood plays a central role in statistical inference. Which of the following best explains what likelihood represents in relation to observed data and hypotheses?

- a) Probability of hypothesis given data
- b) Probability of data given hypothesis
- c) Probability of prior only
- d) Probability of error

Correct Answer: B

Explanation: Likelihood represents how probable the observed data is under a given hypothesis or parameter value.

Example: Example: Probability of observing symptoms if a disease is present.

19. Naïve Bayes classifiers are widely used in real-world applications. Which of the following best describes the type of problems where this algorithm is most effective?

- a) Clustering problems
- b) Classification problems with high-dimensional data
- c) Regression problems
- d) Sorting problems

Correct Answer: B

Explanation: Naïve Bayes is commonly effective for classification, especially with high-dimensional feature spaces.

Example: Example: Email spam classification using thousands of word features.

20. Overfitting is a common issue in machine learning models. Which of the following best describes a situation where a model is considered to be overfitting the training data?

- a) Model performs well on unseen data
- b) Model captures noise instead of pattern
- c) Model is too simple
- d) Model ignores data

Correct Answer: B

Explanation: Overfitting occurs when a model learns noise or specific training examples instead of the general pattern.

Example: Example: Training accuracy is very high but test accuracy is poor.

21. Underfitting occurs when a model fails to capture patterns in data. Which of the following best describes the characteristics of an underfitted model in practice?

- a) High accuracy always
- b) Fails to learn patterns from data
- c) Too complex model
- d) Overtrained model

Correct Answer: B

Explanation: Underfitting occurs when the model is too simple to capture important patterns.

Example: Example: A straight-line model performs poorly on strongly curved data.

22. The EM algorithm is iterative and requires a stopping condition. Which of the following best explains when the algorithm is considered to have converged?

- a) When data changes
- b) When parameters stabilize across iterations
- c) When iterations increase
- d) When accuracy decreases

Correct Answer: B

Explanation: EM is considered converged when its parameters or objective value changes only slightly across iterations.

Example: Example: Cluster means change by less than a chosen tolerance.

23. Bayesian learning combines prior knowledge with observed data. Which of the following best describes how these two components interact during the learning process?

- a) Prior replaces data
- b) Data replaces prior
- c) Both are combined to update belief
- d) Neither is used

Correct Answer: C

Explanation: Bayesian learning combines prior knowledge and observed data to update beliefs.

Example: Example: A prior disease risk is updated using a diagnostic test result.

24. Concept learning involves identifying a suitable hypothesis. Which of the following best explains the main goal of concept learning in machine learning systems?

- a) To reduce computation
- b) To find hypothesis consistent with data
- c) To eliminate features
- d) To increase dataset size

Correct Answer: B

Explanation: Concept learning seeks a hypothesis that is consistent with the available training examples.

Example: Example: Learn a rule that separates accepted and rejected applications.

25. Hypothesis space search is a key component of concept learning. Which of the following best describes how searching through hypothesis spaces helps in selecting an appropriate model?

- a) Random selection of hypothesis
- b) Exploring possible hypotheses to find best fit
- c) Ignoring data patterns
- d) Reducing dataset size

Correct Answer: B

Explanation: Searching the hypothesis space explores possible models to find a suitable hypothesis.

Example: Example: Compare different classification rules and choose one that fits the training data well.

26. In Bayesian inference, normalization ensures probabilities sum to one. What role does the evidence term play in Bayes' Theorem?

- a) It acts as a scaling factor
- b) It removes prior probability
- c) It increases model complexity
- d) It selects hypothesis randomly

Correct Answer: A

Explanation: The evidence term acts as a normalization or scaling factor for the posterior probability.

Example: Example: It ensures the resulting posterior probabilities sum to one.

27. In Naïve Bayes classification, what is the primary reason for using logarithms when computing probabilities?

- a) To increase accuracy
- b) To avoid numerical underflow
- c) To remove dependencies
- d) To simplify dataset

Correct Answer: B

Explanation: Logarithms convert products of very small probabilities into sums and help avoid numerical underflow.

Example: Example: Instead of multiplying hundreds of tiny word probabilities, add their log probabilities.

28. Which of the following best describes conditional probability?

- a) Probability of two events occurring together
- b) Probability of an event given another event has occurred
- c) Probability of independent events
- d) Probability of random variables

Correct Answer: B

Explanation: Conditional probability measures the probability of an event given another event has occurred.

Example: Example: Probability of rain given that dark clouds are observed.

29. Which factor most affects the performance of a Naïve Bayes classifier in real-world datasets?

- a) Number of classes
- b) Independence assumption violation
- c) Dataset size only
- d) Output format

Correct Answer: B

Explanation: Strong violation of the conditional-independence assumption can affect Naïve Bayes performance.

Example: Example: Two features may be strongly dependent and provide redundant information.

30. In Bayesian networks, what ensures that the model remains computationally efficient?

- a) Large dataset
- b) Conditional independence
- c) Continuous variables
- d) High dimensionality

Correct Answer: B

Explanation: Conditional independence reduces the number of probability relationships that must be represented and computed.

Example: Example: Independent variables require fewer conditional probability values.

31. Which of the following best describes the role of prior knowledge in Bayesian learning?

- a) Eliminates need for data
- b) Guides learning before observing data
- c) Reduces accuracy
- d) Replaces hypothesis

Correct Answer: B

Explanation: Prior knowledge provides an initial belief before new observations are incorporated.

Example: Example: A prior belief that a particular disease is rare.

32. In Maximum Likelihood Estimation, what happens when the dataset size increases significantly?

- a) Prior dominates
- b) Likelihood estimation becomes more accurate
- c) Model becomes random
- d) Complexity decreases

Correct Answer: B

Explanation: With more observations, likelihood-based parameter estimates generally become more reliable.

Example: Example: A mean estimated from 10,000 observations is usually more stable than one estimated from 5.

33. Which of the following is a limitation of MLE?

- a) Uses too much prior knowledge
- b) Ignores prior information
- c) Cannot handle data
- d) Always overfits

Correct Answer: B

Explanation: MLE does not incorporate prior information about parameters.

Example: Example: MLE estimates a parameter from observed data without a prior distribution.

34. Which of the following best describes Bayesian estimation compared to MLE?

- a) Uses only likelihood
- b) Uses both prior and likelihood
- c) Ignores uncertainty
- d) Works only for small datasets

Correct Answer: B

Explanation: Bayesian estimation combines prior information with the likelihood from observed data.

Example: Example: Combine a prior mean and the sample mean to obtain a posterior estimate.

35. What is the main advantage of Laplace smoothing in Naïve Bayes?

- a) Removes features
- b) Handles zero probability problem
- c) Reduces dataset size
- d) Increases bias only

Correct Answer: B

Explanation: Laplace smoothing prevents a feature with zero observed count from producing a zero probability.

Example: Example: Add one to word counts so an unseen word does not make the class probability zero.

36. Which of the following best defines a hypothesis in machine learning?

- a) Final output
- b) Assumed model for mapping inputs to outputs
- c) Dataset
- d) Feature vector

Correct Answer: B

Explanation: A hypothesis is an assumed model or rule that maps inputs to outputs.

Example: Example: $y = 2x + 1$ can be a hypothesis.

37. What is the purpose of hypothesis space in learning algorithms?

- a) Store data

- b) Define all possible models
- c) Reduce training time
- d) Increase features

Correct Answer: B

Explanation: The hypothesis space defines the possible models or rules that a learner can consider.

Example: Example: All allowed decision rules form the hypothesis space.

38. Which of the following best explains generalization in machine learning?

- a) Memorizing training data
- b) Performing well on unseen data
- c) Reducing dataset size
- d) Increasing parameters

Correct Answer: B

Explanation: Generalization means the learned model performs well on data it has not seen during training.

Example: Example: A classifier correctly predicts new test samples.

39. What is the effect of noisy data on learning algorithms?

- a) Improves accuracy
- b) Leads to overfitting
- c) Eliminates features
- d) Reduces complexity

Correct Answer: B

Explanation: Noise can cause a model to learn random patterns and overfit the training data.

Example: Example: A classifier memorizes incorrect labels instead of learning the true pattern.

40. In EM algorithm, what is the main challenge during initialization?

- a) Dataset size
- b) Choosing initial parameters
- c) Output format
- d) Removing features

Correct Answer: B

Explanation: Choosing initial parameters is important because poor initialization can lead to a poor local optimum.

Example: Example: Poor initial cluster centers can cause K-means-like mixture fitting to converge to an

undesirable solution.

41. Which of the following best describes convergence in iterative algorithms?

- a) Increasing error
- b) Stabilization of parameters
- c) Random updates
- d) Dataset reduction

Correct Answer: B

Explanation: Convergence means the parameters or objective have stabilized sufficiently.

Example: Example: Parameter changes become smaller than a chosen tolerance.

42. Which of the following is a characteristic of probabilistic models?

- a) Deterministic outputs
- b) Handling uncertainty
- c) Ignoring data
- d) Fixed predictions

Correct Answer: B

Explanation: Probabilistic models explicitly represent uncertainty using probabilities.

Example: Example: A model predicts a 70% chance of rain.

43. What is the role of likelihood in Bayesian inference?

- a) Represents prior
- b) Represents evidence
- c) Measures fit of data to hypothesis
- d) Removes uncertainty

Correct Answer: C

Explanation: Likelihood measures how well the observed data fits a particular hypothesis or parameter value.

Example: Example: Compare which parameter value makes the observed data most likely.

44. Which of the following best describes overfitting?

- a) Model too simple
- b) Model fits noise in training data
- c) Model ignores data
- d) Model has low variance

Correct Answer: B

Explanation: Overfitting occurs when a model fits noise or training-specific details instead of general patterns.

Example: Example: Very high training accuracy but low test accuracy.

45. Which of the following best describes underfitting?

- a) Model too complex
- b) Model fails to capture patterns
- c) Model memorizes data
- d) Model has high variance

Correct Answer: B

Explanation: Underfitting occurs when a model is too simple to capture important patterns.

Example: Example: A simple linear model performs poorly on nonlinear data.

46. What is the role of training data in learning algorithms?

- a) Defines output only
- b) Guides model learning
- c) Reduces complexity
- d) Eliminates hypothesis

Correct Answer: B

Explanation: Training data provides examples from which the model learns parameters or rules.

Example: Example: Student marks and labels are used to train a pass/fail classifier.

47. Which factor improves model generalization?

- a) Increasing noise
- b) Proper regularization
- c) Reducing data
- d) Ignoring validation

Correct Answer: B

Explanation: Proper regularization can reduce overfitting and improve performance on unseen data.

Example: Example: L2 regularization discourages excessively large model weights.

48. Which of the following best explains feature selection?

- a) Removing dataset
- b) Selecting relevant features

- c) Increasing features
- d) Ignoring attributes

Correct Answer: B

Explanation: Feature selection chooses relevant input variables and removes unnecessary ones.

Example: Example: Keep age and income while removing an irrelevant ID column.

49. Which of the following best describes classification problems?

- a) Predict continuous values
- b) Assign labels to data
- c) Cluster data
- d) Reduce dimensions

Correct Answer: B

Explanation: Classification assigns data to discrete categories or labels.

Example: Example: Email → Spam or Not Spam.

50. Which of the following best describes regression problems?

- a) Predict discrete labels
- b) Predict continuous values
- c) Cluster data
- d) Classify data

Correct Answer: B

Explanation: Regression predicts continuous numerical values.

Example: Example: Predicting a house price such as ₹50 lakh.

FINAL ANSWER KEY

1-B, 2-B, 3-C, 4-B, 5-B, 6-A, 7-C, 8-A, 9-B, 10-C 11-B, 12-B,
13-B, 14-B, 15-B, 16-B, 17-B, 18-B, 19-B, 20-B 21-B, 22-B,
23-C, 24-B, 25-B, 26-A, 27-B, 28-B, 29-B, 30-B 31-B, 32-B,
33-B, 34-B, 35-B, 36-B, 37-B, 38-B, 39-B, 40-B 41-B, 42-B,
43-C, 44-B, 45-B, 46-B, 47-B, 48-B, 49-B, 50-B